

SCS 3204: AUTOMATA THEORY

Lecture IX: Non-determinism

Non-determinism

- Non-determinism occurs whenever the outcome of a computation is dependent upon CHOICE
- Regular expressions and transition graphs are nondeterministic
- A nondeterministic automaton allows a string to be accepted by the same machine in more than one way

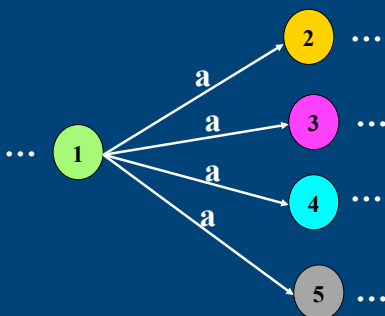
Definition

- A nondeterministic finite automaton (NFA) consists of:
 1. A finite set of states with one start state (-) and some final states (+)
 2. An alphabet Σ of possible input letters
 3. A finite set of transitions that describe how to proceed from each state to other states along edges labelled with letters of the alphabet (**but not Λ**), where we allow more than one edge with the same label from any state and some states where some input letters have no edge.

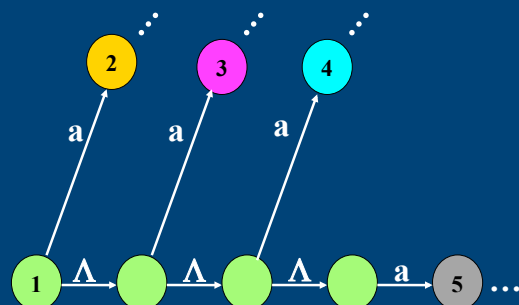
Equivalence

- Note that an NFA is very similar to a TG except that:
 - Its transitions are labelled with alphabet letters not strings, and
 - It does not allow the labelling of transitions with Λ
 - However, any NFA can easily be converted into a TG as we will see...

An NFA with similar labelled edges



But note that it is equivalent to...



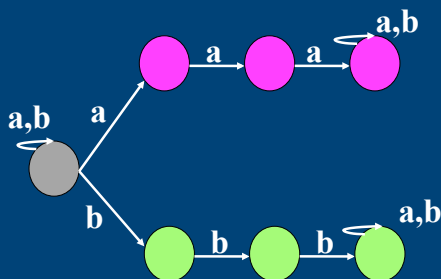
Equivalence Cont.

- From the definition: every DFA is an NFA
- From what we demonstrated in the previous slide: Every NFA has an equivalent TG
- By Kleene's theorem: Every TG has an equivalent DFA – Therefore:
- NFAs define exactly the same set of languages as FAs, TGs and REs!

Why NFA?

- NFA make it much easier to join machines together e.g. $FA_1 + FA_2$, FA_1FA_2
- NFA are easy to read and therefore they can more clearly demonstrate the definition of a language
- e.g. (overleaf) the language of all strings that contain the substring **aaa** or **bbb**.

Example NFA



What is the equivalent TG and DFA?

NFA and Representation

- Non-determinism can have important practical applications especially in AI
- e.g. suppose we want to solve the maze below...

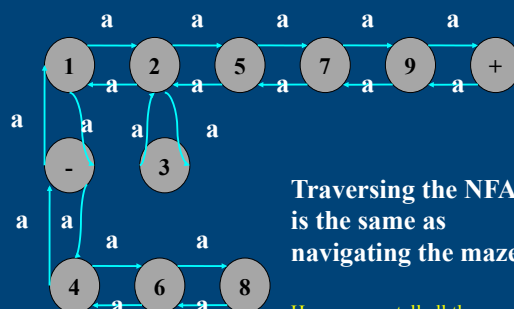
START			
			END

NFA and Representation

- Non-determinism can have important practical applications especially in AI
- e.g. suppose we want to solve the maze below...

START	1	2	3
4		5	
6		7	
8		9	END

NFA and representation cont...



Traversing the NFA is the same as navigating the maze!

How can we tell all the possible solution paths for this maze?

Comparison Table for Automata

	<i>DFA</i>	<i>TG</i>	<i>NFA</i>
<i>Start states</i>			
<i>Final states</i>			
<i>Edge labels</i>			
<i>Edges from states</i>			
<i>Deterministic?</i>			
<i>1 word 1 path?</i>			

Exercises

- Develop nondeterministic finite automata for the following languages defined over the alphabet $\Sigma = \{a,b\}$
 - aaa
 - $(a + b)^*$
 - $b(a + b)^*$
 - $a(a + b)^*b + b(a + b)^*a$
 - All strings in L_A , where $L_A = \{x \in \{a,b\}^* : |x| \geq 3, \text{ and } 3^{rd} \text{ symbol of } x \text{ from the right is 'a'}\}$
 - All strings over $\{0,1\}$ which end in 1 but do not contain 00